

Practice Quiz 05A — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 05

Contraposition and a universally quantified goal.

Recall that $d \mid n$ means $n = dk$ for some integer k .

1. Prove

$$\forall x \in \mathbf{Z} \left(4 \nmid x^2 \longrightarrow 2 \nmid x \right).$$

Introduce the arbitrary integer explicitly and name your proof strategy.

Solution Let x be an arbitrary integer. We prove the contrapositive: if $2 \mid x$, then $4 \mid x^2$. Suppose $2 \mid x$. By definition, $x = 2k$ for some integer k . Therefore

$$x^2 = (2k)^2 = 4k^2.$$

Since k^2 is an integer, $4 \mid x^2$. This proves the contrapositive, and hence $4 \nmid x^2 \longrightarrow 2 \nmid x$. Since x was arbitrary, the universally quantified statement follows. ■